Reg. No. | | | | | | | | | | | [14ME72]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2019

OPERATIONS RESEARCH

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer any **FIVE** full Questions choosing atleast two from each part.

ii. Missing data may be suitably assumed.

PART - A

1. a. Write the characteristics of operations research. **04**

b. A person wants to decide the constituents of a diet which will fulfill his daily requirements of proteins, fats and carbohydrates at the minimum cost. The choice is to be made from four different types of foods. The yields per unit of these foods are given below; **10**

Food type	Yield per unit			Cost per unit
	Proteins	Fats	Carbohydrates	
1	3	2	6	45
2	4	2	4	40
3	8	7	7	85
4	6	5	4	65
Minimum requirement	800	200	700	

Formulate linear programming model for the problem.

2. a. Explain alternate optimal solution with relevant example. **05**

b. Use Big- M simplex method to solve the following LPP: **09**

Maximize $Z = X_1 + 2X_2 + 3X_3 - X_4$

Subject to: $X_1 + 2X_2 + 3X_3 = 15$

$2X_1 + X_2 + 5X_3 = 20$

$X_1 + 2X_2 + X_3 + X_4 = 4$ $\quad X_1, X_2, X_3, X_4 >= 0$

3. a. Briefly narrate different methods of basic feasible solution, & optimal solution methods. **04**

b. A company has three factories at A, B, C which supply ware houses D, E, F, G and H. Monthly capacities are 19, 28, and 25 respectively, requirements are 11, 13, 7, 17, and 24 respectively, **unit shipping cost** are follows, located in three cities viz. X, Y, Z. These factories ; **10**

	D	E	F	G	H
A	42	42	44	40	44
B	34	42	40	46	48
C	46	44	42	48	46

Find the optimal cost for the same.

4. a. Distinguish between assignment and transportation problem. **04**

b. 5 jobs are to be processed and five machines are available. Any M/c can process any job with the **profit** in the table as follows. Determine the optimal assignment condition. **10**

		Machines				
		A	B	C	D	E
Jobs	1	32	38	40	28	40
	2	40	24	28	21	36
	3	41	27	33	30	37
	4	22	38	41	36	36
	5	29	33	40	35	39

PART - B

5. a. In a factory, there are six jobs to process, each of which should go to machines A & B in the order AB. The processing timing in minutes are given, determine the optimal sequencing & total elapsed time. **07**

Jobs	1	2	3	4	5	6
Machine A	7	4	2	5	9	8
Machine B	3	8	6	6	4	1

b. A machine operator has to perform 3 operations, turning, threading & knurling on three machines A, B & C in the order ABC. Find the optimum sequences when the time in hours are given. **07**

Job	Turning A_i	Threading B_i	Knurling C_i
1	3	8	13
2	12	6	14
3	5	4	9
4	2	6	12
5	9	3	8
6	11	1	13

6. a. Solve the game by pay off matrix. **06**

	Player A		
Player B	5	0	2
	-1	8	6
	1	2	3

b. Solve the following game, whose pay off matrix is given below. **08**

	Player B		
Player A	0	-2	7
	2	5	6
	3	-3	8

7. a. List the differences between PERT and CPM with application. **05**

b. An R &D activity has seven activities for which three time estimates are given below along with the precedence activities. **09**

Activity	Precedence activity	Optimistic time	Most likely time	Pessimistic time
A	None	4	6	8
B	A	6	10	12
C	A	8	18	24
D	B	9	9	9
E	C	10	14	18
F	A	5	5	5
G	D,E,F	8	10	12

i. Draw the PERT network.

ii. Draw the critical path with EST,LST and expected project duration.

iii. What is the probability that the project will be completed not more than 3 weeks later than expected?

8. a. With suitable examples briefly explain the following with respect to queuing theory. **05**

i. Balking

ii. Jockeying

iii. Reneging

b. In a hair dressing salon with one barber, the customer service follows a Poisson distribution at an average rate of one every 45 minutes, the service time is exponentially distributed with a mean of 30 minutes, find **09**

i. Average number of customers in the saloon

ii. Average waiting times of a customers in a saloon,

iii. Average idle time of the barber.